PS02 - My Answers (Quants 1)

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Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in `R`).

`Answer:`

Okay, before we can start, we need to actually 'import' this table into RStudio. So let's create our dataframe: `R Code:`

```
#Option A
data_file <- c(14, 6, 7, 7, 7, 1)
df <- matrix(data = data_file , nrow = 2, ncol = 3, byrow =TRUE)
rownames(df) <- c("upper class", "lower class")
colnames(df) <- c("Not Stopped", "Bribe requested", "Stopped/given
    warning")

print(df)

#Option B
Option_B_df <- matrix(NA, nrow= 2, ncol=3)
rownames(Option_B_df) <- c("upper class", "lower class")
colnames(Option_B_df) <- c("Not Stopped", "Bribe requested", "Stopped/
    given warning")
Option_B_df[1,] <- c(14,6,7)
Option_B_df[2,] <- c(7,7,1)
print(Option_B_df)
```

Please note: I tried several options of how to effectively create this contingency table. Since Option A has a more suitable name, we will continue with this one.

Now, let's start with the actual task.

There is a very simple option: R can run a Chi^2 test. This creates a dataframe that actually stores a lot of useful information, including the (standardised) residuals.

`R Code:`

```
chi_square_by_R <- chisq.test(df)
print(chi_square_by_R)
```

```
Warning message:
In chisq.test(df) : Chi-squared approximation may be incorrect

Pearson's Chi-squared test
data:  df
X-squared = 3.7912, df = 2, p-value =
0.1502
```

So , the Chi^2 statistic is 3.7912. Moreover, R tells us that it used two degrees of freedom. This is because the $df = (row - 1) * (column - 1)$. Moreover, it also gives us a p-value, which we will return to in a subsequent task.

Importantly, in the console, we also see that R warns us that the approximation may be incorrect. This is because one of the cells contains a value smaller than 5.

Additionally, we can calculate this "by hand". Let's use the dataframe from before. Importantly, the Chi^2 is calculated like this: $\sum(\frac{(f_{observed} - f_{expected})^2}{f_{expected}})$. Thus, we also need to calculate the expected values for each cell. To do this, we calculate: $f_{expected} = \frac{row\ total * column\ total}{grand\ total}$, for each cell.

The following code shows that entire process, culminating in the Chi^2

`R Code:`

```r
#the Chi^2 statistic = the sum of: (observed-expected)^2/expected.
#Thus we first calculate the expected values for each cell.

#To do this, we need the row/column/grand total: expected = ((row total)
    +(column total))/grand total

c_total <- colSums(df)
r_total <- rowSums(df)
grand_total <- sum(df)
#lets check
c_total
r_total
grand_total

#okay lets continue:
#f(expected) =(rowtotal*columntotal)/grandtotal
#lets first figure out how to calculate all the products we need at once
?outer
outer(c_total, r_total) #gives the outer product of the arrays X and Y

#okay, now that I know this works, lets get the expected values for each
    cell all at once
expected_values <- outer(r_total, c_total)/grand_total
#make sure the r_total is first, or the form is wrong

#now let's do the actual Chi^2 statistic
#our elements:
#observed values: in the dataframe df
#expected values: in expected_values

chi_sq_by_hand <- sum((df-expected_values)^2/expected_values)

#okay lets compare
chi_sq_by_hand
chi_square_by_R
```

```
#it's the same. thats good.
```

```
> chi_sq_by_hand
[1] 3.791168

> chi_square_by_R
Pearson's Chi-squared test
data:  df
X-squared = 3.7912, df = 2, p-value =
0.1502
```

(b) Now calculate the p-value from the test statistic you just created (in `R`).[2] What do you conclude if $\alpha = 0.1$?
`Answer:` Okay, let's return to the p-value that we got from R when we ran the Chi^2 test: R shows us a p-value of roughly 0.15. Since 0.15 is bigger than 0.1, we fail to reject the H_0 that the two variables are independent. There is not sufficient evidence based on our sample to accept the alternative Hypothesis that they are dependent.

Once again, we can also do this "by hand". For this, we need the degrees of freedom, which, as mentioned above, are calculated as follows: $df = (row - 1) * (column - 1)$. We then compare our TS to a critical value on the Chi^2 distribution.

`R Code:`

```
#lets calculate the p-value
#first we need the degrees of freedom

#df = (rows1)(columns1)
rows <- nrow(df)
columns <- ncol(df)

df <- (rows - 1) * (columns - 1)

p_value_chi_sq <- pchisq(chi_sq_by_hand, df = df, lower.tail = FALSE)
p_value_chi_sq

#as we can see the p-value is equal to roughly 0.15. Thus, since alpha =
    0.1 and 0.15 > 0.05, we reject the Nullhypotheses that these two
    variables are independent
```

```
> p_value_chi_sq
[1] 0.1502306
```

[2] Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.
`Answer:`

Okay, as mentioned before, the dataframe produced by R gives us a lot of useful info. This includes the standardised residuals.

`R Code:`

```r
chi_square_by_R <- chisq.test(df)
print(chi_square_by_R)
```

```
> print(std_residuals_by_R)
Not Stopped Bribe requested Stopped/given warning
upper class   0.3220306       -1.641957              1.523026
lower class  -0.3220306        1.641957             -1.523026
```

`Now, let's insert them:`

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.322	-1.642	1.523
Lower class	-0.322	1.642	-1.523

Table 1: Standardized residuals by class and outcome

(d) How might the standardized residuals help you interpret the results?

`Answer:`

The standardized residuals can help us identify which cells or groups drive the overall difference observed in the data. Cells with larger standardized residuals contribute more heavily to the χ^2 statistic. If the residual is positive, it means this outcome was observed more often than expected under the Nullhypothesis. Conversely, if the residuals are negative, we actually observed a certain outcome less often than expected under the Null.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
`GP`	An identifier for the Gram Panchayat (GP)
`village`	identifier for each village
`reserved`	binary variable indicating whether the GP was reserved for women leaders or not
`female`	binary variable indicating whether the GP had a female leader or not
`irrigation`	variable measuring the number of new or repaired irrigation facilities in the village since the reserve policy started
`water`	variable measuring the number of new or repaired drinking-water facilities in the village since the reserve policy started

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis. `Answer:`

So, as stated above, we need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Thus, plainly speaking, the H_0 is that the reservation policy has *no* impact on the number of new or repaired drinking water facilities in the villages. This corresponds to $H_0 : \beta_1 = 0$.

The H_a is that the reservation policy does have an impact on the number of new or repaired drinking water facilities in the villages. This corresponds to $H_a : \beta_1 \neq 0$. *Technically, given the description, the hypothesis would be that it has a positive effect, but it is a two-tailed hypothesis test, so it is AN effect.*

(b) Run a bivariate regression to test this hypothesis in `R` (include your code!).

While it is a bit naive, we're just gonna run a very simple regression model, where we regress the outcome variable on our predictor.

`R Code:`

```r
#simple bivariate regression:
reg_1 <- lm(data$water ~ data$reserved)
summary(reg_1)
```

```
> summary(reg_1)

Call:
lm(formula = data$water ~ data$reserved)

Residuals:
Min      1Q  Median      3Q     Max
-23.991 -14.738  -7.865   2.262 316.009

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)     14.738      2.286   6.446 4.22e-10 ***
data$reserved    9.252      3.948   2.344   0.0197 *
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.45 on 320 degrees of freedom
Multiple R-squared:  0.01688, Adjusted R-squared:  0.0138
F-statistic: 5.493 on 1 and 320 DF,  p-value: 0.0197
```

(c) Interpret the coefficient estimate for reservation policy.

`Answer:`

Okay, as a last step, let's interpret the output of this regression.

The slope coefficient in a model is the (average) expected change in the outcome variable for a one-unit increase in the predictor. In this case, the predictor is binary.

Our regression shows that based on our sample there is evidence that the reservation policy has an effect on the number of new or repaired drinking water facilities in villages (thus, we reject $H_0 : \beta_1 = 0$). Specifically, we find that, on average, villages with reserved seats for women have 9.25 more new or repaired drinking water facilities than villages without such a policy.

The intercept ($\beta_0 = 14.74$) tells us that in villages without a reservation policy, the expected number of new or repaired drinking water facilities is roughly 15.

It is important to note that β_1 is an estimator with its own sampling distribution. Based on our sample, we can calculate a 95% confidence interval to make the uncertainty around this estimate more apparent:

`R Code:`

```r
confint(reg_1, level =0.95)
```

```
> confint(reg_1, level =0.95)
2.5 %   97.5 %
(Intercept)   10.240240 19.23640
data$reserved  1.485608 17.01924
```

The resulting 95% confidence interval is approximately [1.49, 17.02], meaning we are 95 % confident that the true effect of the reservation policy on water infrastructure lies within this range. Since the interval does not include zero, the effect is statistically significant at the level p = 0.05

Hence, we conclude that the presence of reserved seats for women is associated with an increase in the number of new or repaired drinking water facilities in villages.